

reduced. Remember that the X1900XT is a mighty 48-pixel unit core, and that alone gives it awesome Shader performance, especially at higher resolutions as compared to the other Radeons.

We decided to bump up the speeds ourselves to the ATI-recommended values, just to see what would happen. We found that these cards are neck-and-neck with the 7900GTXs, but increased core temperatures by another 12 degrees! Since most users would not overclock cards themselves, we decided to log all scores using only what the manufacturers sent us.

ATI has also recently launched a new flagship model—the X1950XTX—which also marks the debut of GDDR4 memory in the mainstream graphics market. This memory is rated at 2000 MHz (1000 MHz DDR), and we're told that GDDR4 could scale all the way to 2.6 GHz, which would do wonders for today's (and tomorrow's) bandwidth-starved GPUs. Unfortunately, the X1950XTX was launched around the same time as this test, and we couldn't get our hands on a test sample in time. Look forward to it in *Bazaar* in the coming months though.

A Shade(r) Lighter NVIDIA 7900GT, 7800GTX ATI X1900GT, X1800XL, X1800GTO

These cards represent the next best thing from the cards in the previous category, though they're still high-end. What detracts from performance adds value for money. Thirteen cards fell in this category.

Features

NVIDIA's 7900GT and 7800GTX share the same 24-pixel pipe architecture, though the former is a newer product, and foregoes some of the latter's transistor count in favour of ramped-up clock speeds. The cooling solution on the 7900 is puny indeed, but it does the job reasonably well—all 7900GT based cards ran stably during

resolution is called Anti-Aliasing. In 3D textures, Aliasing is most visible when two textures with differing qualities overlap. For example, a wall and a door, or even a rock wall, with a lot of Bump Mapping. It gets worse as the viewing perspective changes, and the wall may even seem to be moving!

HDR (High Dynamic Range): Dynamic Range refers to the variation or contrast to the light and dark areas in a given scene. The purpose of High Dynamic Range lighting is to accurately represent the varying intensity of light accurately, from bright sunshine glinting off, say, a gun barrel or a window pane, to variations in intensity of shadows. The core objective behind HDR rendering techniques is, of course, visual realism. Enabling HDR requires a card that supports Shader Model 2 or 3, and the performance hit can be quite significant depending, of course, on the variation in lighting in the given 3D scene.

Soft Shadows: Soft Shadows are, as the name suggests, shadows that are created from multiple light sources of varying intensities. In games, shadows are typically just that—dark and gloomy. Soft Shadows introduces the concept of creating shadows that may be dark, or slightly greyish, or even lighter. This adds realism to a scene, making the environment seem much more gritty and believable. Soft Shadows, as of now, has yet to see proper implementation in games. *F.E.A.R.* is a title that has heavily utilised soft shadows—unfortunately, the visual effects are minimal in comparison to the colossal drop in frames rendered per second.

Lighter Chipset features and specs

	NVIDIA		ATI		
Part Name	7900GT	7800GTX	X1900GT	X1800XL	X1800GTO
Chipset Designation	G71	G70	R580	R520	R520
Manufacturing Process (nm)	90	110	90	90	90
Core Clock Speed (MHz)	450	430	500	400	400
(Manufacturer Reference Speeds)					
Onboard Memory (MB) / Type	256 / GDDR3	256 / GDDR3	256 / GDDR3	256 / GDDR3	256 / GDDR3
(Manufacturer reference)					
Memory Speed (MHz)	1320	1200	1200	1000	1000
(Manufacturer reference speeds)					
Memory bus width (Bits)	256	256	256	256	256
Pixel Shader units	24	24	36	16	12
Vertex Shader Units	8	8	8	8	8

our tests. The 7800GTX cards feature a single-slot cooler, but it's much more robust—which is needed because they heat up more as well. The Gigabyte 7900GT featured a Zalman cooler that looked... umm... cool! Part of the core was visible on this card, and the cooler was really light.

Another card that shipped with a custom cooler was the ASUS Extreme N7800GTX Top. Based around an overclocked 7800GTX core, with 256 MB of memory onboard, this card looked good with its airplane propeller-like cooling fan and massive heatsink. The heatsink-fan combo look a lot like a high performance Arctic NV Silencer—for those not in the know, Arctic makes some of the best VGA cooling solutions.

ATI also recommends a single-slot cooler for their cores under this category, as these cards don't run as hot. It's hard to ignore the ATI All-In-Wonder X1800XL when it comes to looks, performance, and features. ATI's TV-Tuners are some of the best around, and this card will game ferociously and record all your favourite shows with equal aplomb! Naturally, the card was accompanied with all the expected paraphernalia—including a feature-packed remote and an FM radio antenna.

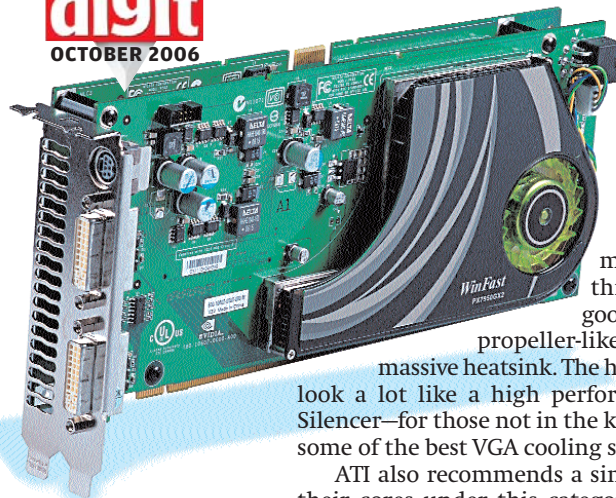
The Powercolor X1900GT, too, looked attractive, mainly because a large ATI Ruby adorns the metal fan-cum-heatsink cover. Incidentally, the GT version of the Radeon X1900 cards sport a single-slot cooler. That doesn't mean that the X1900GT heats up any less though—quite the contrary; we noted temperatures in the high seventies on the Powercolor card, which suggests that with prolonged gaming, temperatures could easily approach "XT" territory!

The bundle accompanying the ASUS Extreme N7800GTX TOP just deserves a quick word here—simply awesome: 11 discs with four full-version games. The Sapphire cards had very skimpy bundles as far as CD/DVDs go, but came with all the connectors you'll ever need!

Performance

The factory-overclocked Gigabyte 7900GT performed well through all the tests in this category. It was closely dogged by another overclocked

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Leadtek PX7950GX2
The Hot one